



<Intelligent Traffic Cone Control and Deployment System>

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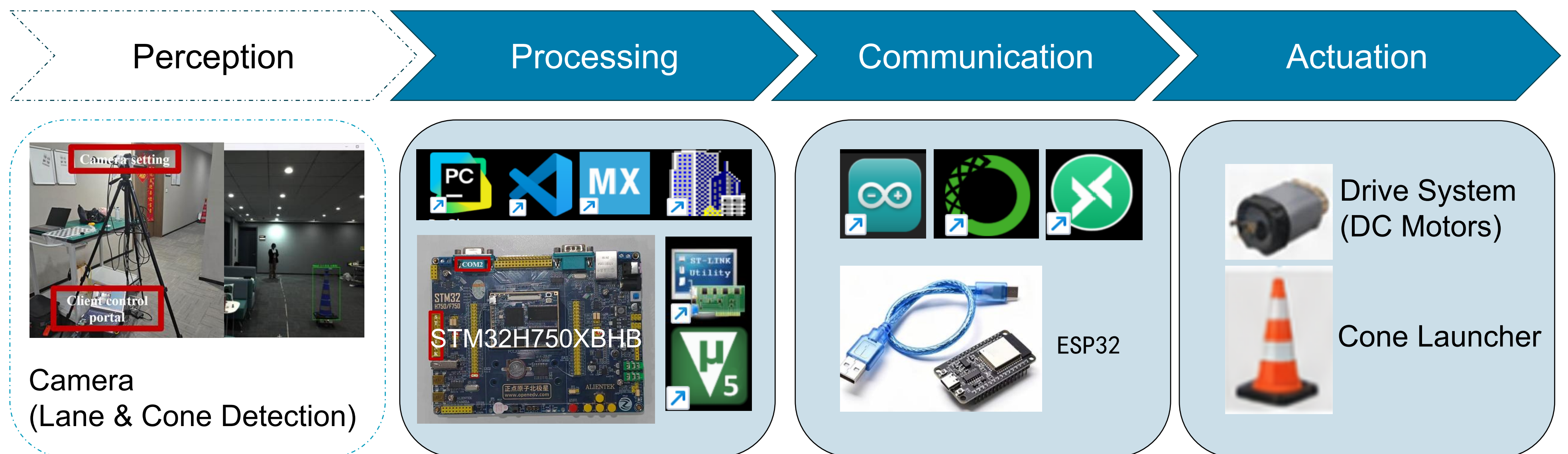
1. INTRODUCTION

Roadworks and lane closures are major causes of traffic congestion and secondary-accident risk. Manual cone deployment is labor-intensive, inconsistent, and unsafe. This project proposes an intelligent system that automates traffic cone deployment and control using computer vision, robotics, and IoT system to improve efficiency, safety, and adaptability.

2. OBJECTIVES

- Develop an autonomous cone deployment robot
- Detect cones and lane boundaries using vision
- Enable remote monitoring and manual override
- Evaluate system performance in real-world scenarios

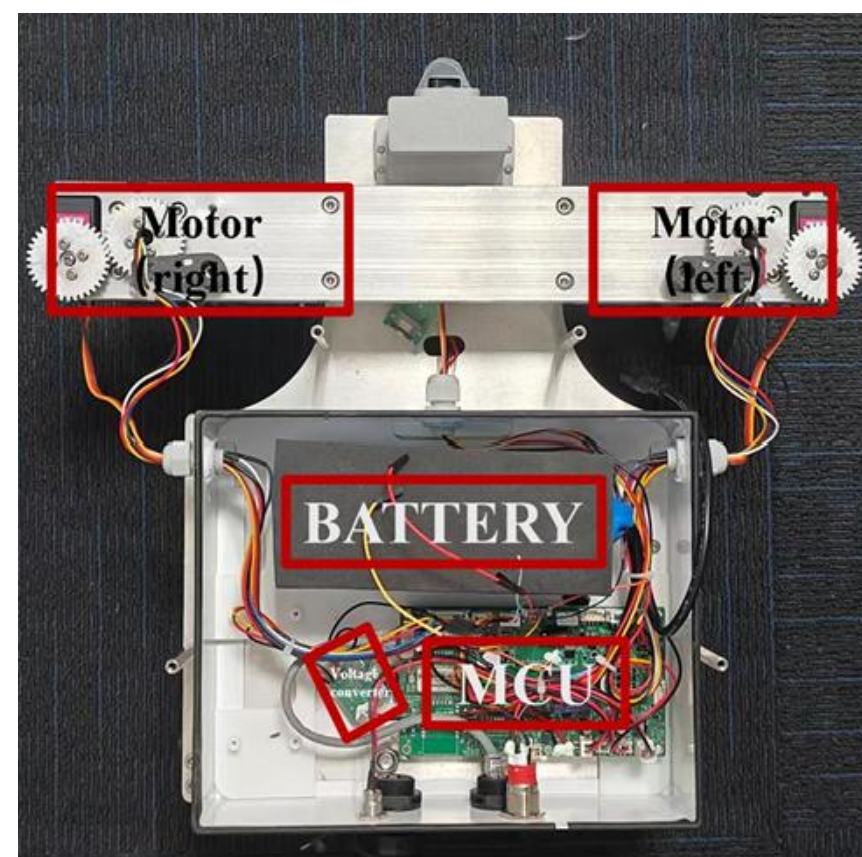
3. SYSTEM OVERVIEW



5. SYSTEM IMPLEMENTATION



Upper body
(Vision recognition)



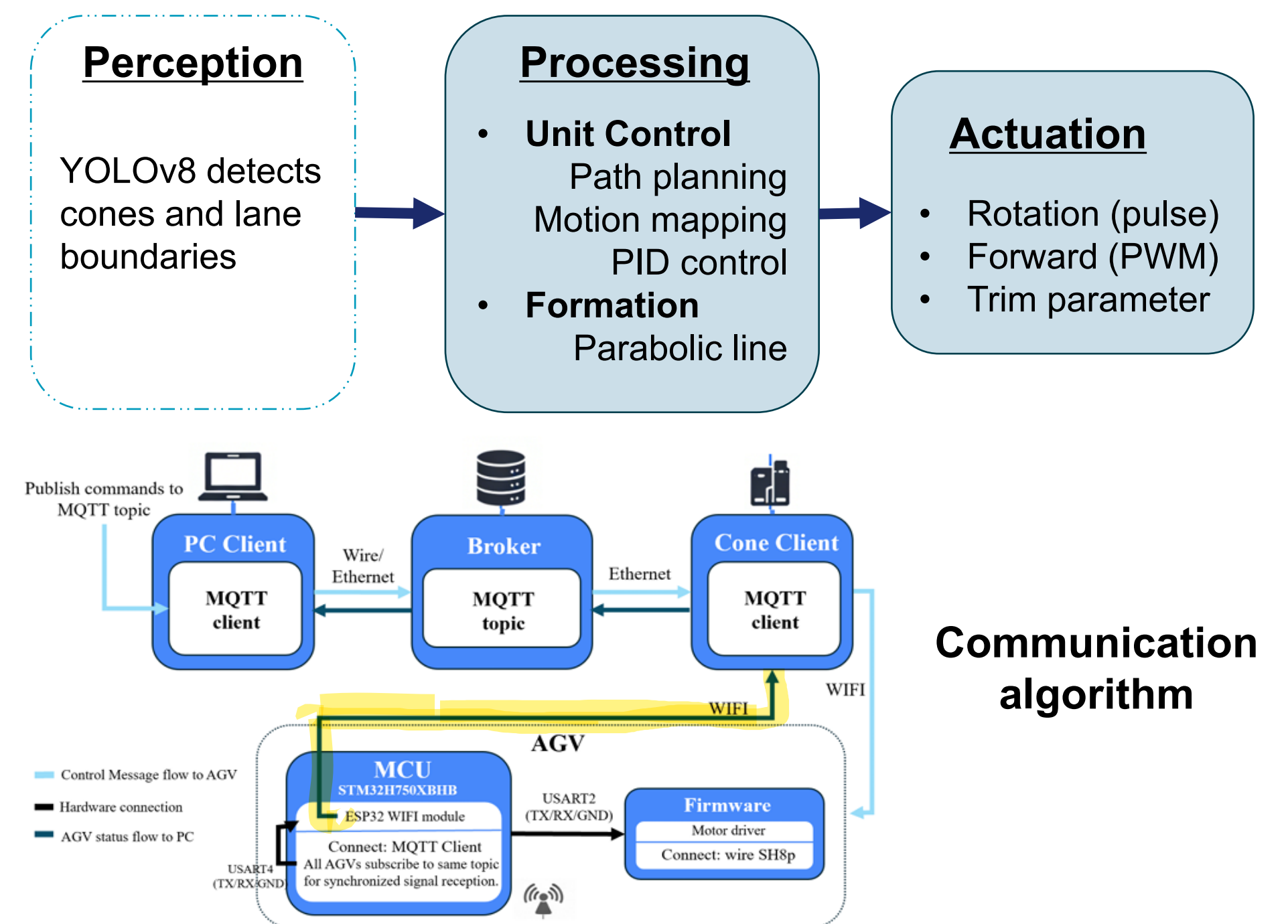
Lower body
(Control algorithm)

7. FUTURE WORK

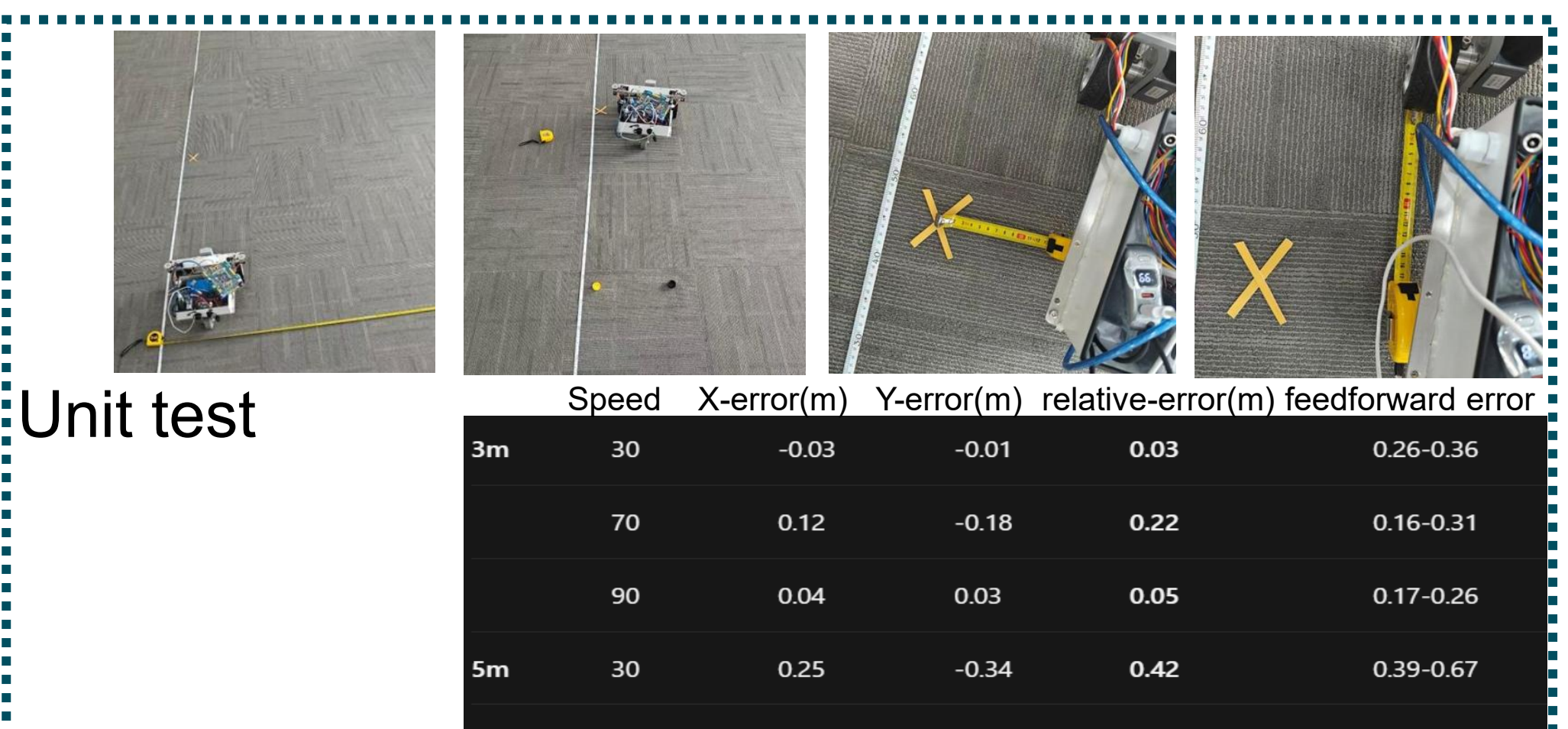
- Multi-robot coordination testing
- Improved obstacle avoidance and path optimization
- Extended battery life and weatherproof design

4. METHODOLOGY

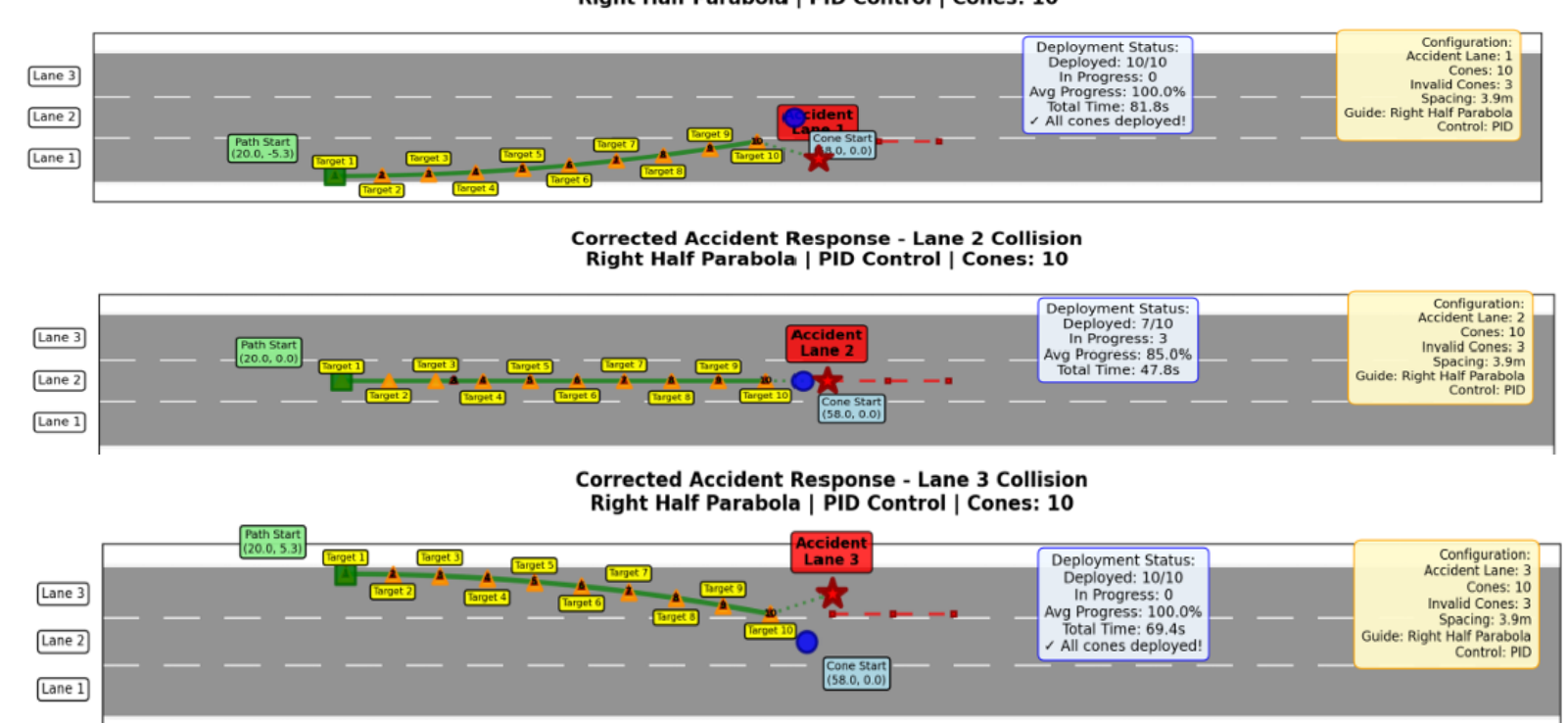
The system integrates perception, decision-making, actuation, and communication to achieve intelligent traffic cone control.



6. RESULTS



Formation test



ACKNOWLEDGEMENT